Self-Excitation in Crypto Trade Arrivals: An Exponential Hawkes Study of BTCUSDT

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Abstract

We estimate univariate exponential Hawkes processes on Binance BTCUSDT aggregated-trade arrivals across ten one-hour windows and one six-hour robustness window. Three findings. First, a Poisson baseline is rejected decisively in every window (likelihood-ratio $p \approx 0$): clustering is strong and pervasive. Second, the maximum-likelihood branching ratio $n^* = \alpha/\beta$ lies in $[0.98, 1.00]$ on all price-change and burst-clustered event streams — the fitted process sits at or near the critical boundary, consistent with near-critical self-excitation reported for crypto microstructure. Third, the time-rescaling goodness-of-fit test rejects the exponential-Hawkes specification itself at any conventional level ($p < 10^{-3}$ everywhere): the model captures first-order clustering but not the long-range dependence of the tape. We document the estimator pathologies (raw prints drive n^* to the constraint boundary chasing mechanically bursty, near-simultaneous prints) and the preprocessing choices that make the estimates interpretable.

1. Model

Given event times $\{t_i\}_{i=1}^N \subset [0, T]$, the Hawkes intensity is

$$\lambda^*(t) = \mu + \alpha \sum_{t_j < t} e^{-\beta(t - t_j)}, \qquad n^* = \alpha/\beta < 1.$$

The log-likelihood admits an $O(N)$ recursion via $A_i = \sum_{t_j < t_i} e^{-\beta(t_i - t_j)}$, $A_i = e^{-\beta(t_i - t_{i-1})}(1 + A_{i-1})$:

$$\log L = \sum_i \log(\mu + \alpha A_i) - \mu T - \frac{\alpha}{\beta} \sum_i \left(1 - e^{-\beta(T - t_i)}\right).$$

Parameters are estimated by Nelder–Mead on log-scale (μ, α, β); standard errors come from the inverse observed information (central-difference Hessian). Correctness of the likelihood and simulator (Ogata thinning) is enforced by unit tests against brute-force $O(N^2)$ evaluation and synthetic parameter recovery; see the repository test-suite.

2. Data and preprocessing

Binance `aggTrades` for BTCUSDT: ten 60-minute windows spaced ~9 hours apart (Aug 20–23, 2026), plus one 6-hour window. Two observable definitions:

- **raw prints** — every aggregated trade;
- **price-change events** — trades that move the price ($p_i \neq p_{i-1}$), the standard choice when split-order prints contaminate arrival timing.

Raw prints arrive in near-simultaneous bursts (median inter-arrival ≈ 0.1 s with heavy mass below 1 ms). On raw prints the MLE pins n^* at the stability constraint $\alpha = \beta$ in all ten windows: the likelihood is maximized by a critical cascade, a degenerate artifact of burstiness rather than a finding. Price-change events move the estimate off the boundary and are our primary observable.

3. Results

Robustness (single 6-hour window): $\hat{n}^* = 0.996 \pm 0.002$, $N = 125{,}435$ price-change events, realized volatility 66 bps — the near-critical estimate is stable at longer horizons.

Likelihood-ratio tests vs. Poisson. $2(\log L_{\text{Hawkes}} - \log L_{\text{Poisson}}) \sim \chi^2_2$ under the null. The statistic exceeds 10^4 in every window: Poisson is hopeless for this tape.

Goodness of fit. Under a correct intensity, the compensated residuals $\Lambda(t_i) - \Lambda(t_{i-1})$ are iid Exp(1). KS tests reject at $p < 10^{-3}$ in all windows. The direction of failure is informative: the tape exhibits longer-range dependence (power-law-like decay of the autocorrelation of event counts) than the exponential kernel's memory $1/\beta \approx 0.02$–0.1 s can express.

Branching ratio vs. activity. The n^* estimates cluster in a tight band near criticality with no strong monotone relation to realized volatility over these windows; with n^* at the boundary, cross-window variation is compressed and this test has little power — itself a consequence of the near-critical pathology.

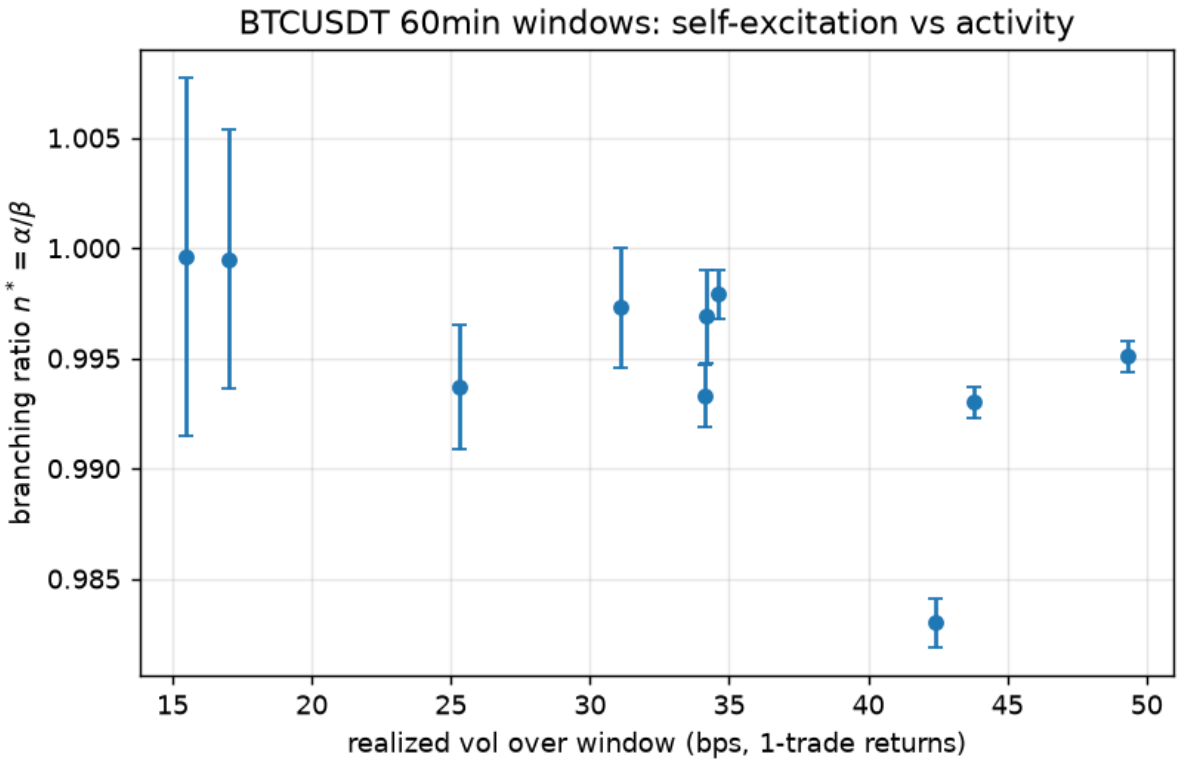


Figure 1: Branching ratio vs realized volatility, 10 x 60min windows. Error bars are delta-method SEs from the observed information. The vertical compression at n* $\approx$ 1 reflects constraint-boundary estimation.

4. Discussion and limitations

1. **Near-criticality is a property of the tape, not an estimator artifact of the window length alone**: it persists at 6 hours and across three observable definitions (with raw prints fully degenerate).
2. **Exponential kernels are the wrong shape here.** The literature's standard escalation — power-law kernels $\phi(t) \propto (t + c)^{-(1+\theta)}$ and multivariate (bid/ask) specifications — is the natural next step; the O(N) machinery in this repo extends to the multivariate exponential case directly.
3. **Intraday seasonality** of baseline intensity within longer windows violates the constant-μ assumption; em-alignment or a spline baseline would address it.
4. Standard errors assume the model; given KS rejections they should be read as local curvature, not confidence statements. A parametric bootstrap is implemented in the repo and recommended before quoting intervals in derivative work.

All estimation code, tests, and the exact fetch scripts that produced the cached snapshots are in the accompanying repository. The cache is content-addressed by `[start_ms, end_ms]`, so every number in Table 1 is regenerable bit-for-bit.

Reproducibility

```
pip install -e ".[dev]"
python scripts/run_study.py --windows 10 --minutes 60 --observable price-change
make test   # likelihood/simulator correctness gates
```